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Vector Search Best Practices — Embedding & Retrieval Guide

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> This guide consolidates NovaTech's hard-won lessons from deploying vector search at scale for Project Atlas. It covers embedding model selection, chunking strategies, HNSW index tuning, hybrid retrieval design, and evaluation metrics. Teams using Qdrant must review this guide before creating new collections.

1. Embedding Model Selection

After evaluating 8 embedding models on our internal benchmark (3 200 query-document pairs), BAAI/bge-base-en-v1.5 (768 dims) was selected as the standard dense model. It achieved the best NDCG@10 score (0.74) while remaining deployable on CPU for batch inference.

Model	NDCG@10	Latency (ms)	Dims	Decision
BAAI/bge-base-en-v1.5	0.74	45	768	Selected
text-embedding-3-small	0.72	110	1536	Rejected (cost)
all-MiniLM-L6-v2	0.61	12	384	Rejected (quality)
BAAI/bge-large-en-v1.5	0.76	210	1024	Rejected (latency)

2. Chunking Strategy

Documents are chunked using a recursive character splitter with chunk_size=512 tokens, chunk_overlap=64 tokens. Sentence boundaries are respected via a sentence-aware split that never breaks mid-sentence. Metadata (title, section heading, page number, classification) is stored as Qdrant payload alongside each chunk.

- Code blocks are chunked separately with a max size of 1 024 tokens.
- Tables are extracted as standalone chunks with a table prefix in the text.
- Short chunks < 50 tokens are merged with the next sibling chunk.
- Parent-child chunking is used for context window expansion in reranking.

3. HNSW Index Tuning

Production Qdrant collections use m=16, ef_construct=200 for index build, and ef=128 for search. These values were selected via a grid search balancing recall@10 (>0.97 target) and memory usage. Increasing m beyond 16 improved recall by only 0.4% while doubling index size.

4. Evaluation Framework

Retrieval quality is measured weekly using RAGAS: faithfulness, answer relevance, context precision, and context recall. Scores are tracked in a Grafana panel. A drop of more than 0.05 in any metric over 7 days triggers a retrieval review meeting.